

Supplementary Material for “Policy-to-Purpose-to-Byte Continuity for Post-Quantum Peer-to-Peer Sessions”

Zi’ang Li

I. SCOPE

This supplement is intentionally subordinate to the main manuscript. It will hold detailed encodings, derivations of theorems stated in the main text, extended evidence tables, and reproduction instructions. It will not introduce an unreferenced theorem, a new protocol, or a separate contribution track.

II. NORMATIVE ENCODING INVENTORY

The final version will give field identifiers, canonical order, type, length bound, validation rule, transcript projection, model symbol, implementation symbol, and negative vector for every field in `AuthenticatedPolicySnapshot`, each endpoint’s `LocalAuthenticatedDecision`, its peer-visible `wireDecisionDigest` projection, `LocalEffectScopeSet`, `BoundSessionContextV1`, `Message A`, `Message B`, `Finished`, `GrantReadyV1`, `sharedGrantId`, `bilateralReadyDigest`, `abort/revoke` messages, the opaque grant handle and service record, each typed trusted-effect outcome, journal state, and purpose-specific receipt.

TABLE I

EXAMPLE TRACEABILITY COLUMNS. VALUES REMAIN PENDING UNTIL THE NORMATIVE CODEC EXISTS.

Object	Specification	Formal event	Code/evidence
Policy snapshot	Pending	Pending	Pending
Local decision	Pending	Pending	Pending
Effect scope	Pending	Pending	Pending
Purpose context	Pending	Pending	Pending
Owner transaction	Pending	Pending	Pending
Ready set	Pending	Pending	Pending
Session grant	Pending	Pending	Pending
Trusted effect	Pending	Pending	Pending
Effect receipt	Pending	Pending	Pending

III. FORMAL ARTIFACT INVENTORY

For every main-text property, the final supplement will list the exact theorem or lemma name, model revision, tool and version, assumptions, compromise escapes, executability lemma, negative control, report digest, and the main-text claim it supports. EasyCrypt, Tamarin, ProVerif, and the state

model will remain separate rows because their abstractions and conclusions differ.

IV. EXTENDED EVALUATION MATRIX

The final matrix will identify experimental unit, endpoint and binary identity, transport, purpose, suite, policy, attack or ablation, batch structure, primary metric, confidence interval, raw-data path, and generator for each reported result. Static checks, simulator runs, physical devices, bare-metal hosts, and published artifacts will be labeled as different evidence classes.

V. REPRODUCTION

The review artifact will provide one entry point that validates the claim ledger and manifest, rebuilds formal reports, compiles the normative core and adapters in isolated checkouts, runs the bounded portable tests, and regenerates every submitted table and figure from raw inputs. Hardware-dependent lanes will have separate commands and will not be reported as reproduced when only their historical transcripts are present.

VI. OPEN EVIDENCE ITEMS

At this draft stage, the integrated codec, formal theories, state model, bilateral-ready implementation, trusted effect committers, cross-language corpus, final binaries, physical-device sessions, non-Apple interoperability, application receipts, raw performance data, and immutable evidence manifest are all pending. This section will be removed only as each item is replaced by source-bound evidence and the claim ledger validates the state transition.